

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November-December 2012

CL203.01 Fluid Mechanics - I

Date: 29.11.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Define Co-efficient of dynamic viscosity and Weight density [02]
- (b) Determine the specific gravity of a fluid having viscosity 0.07 poise and kinematic viscosity 0.042 stokes. [02]
- (c) The horizontal flat plates are placed 0.15 mm apart and the space between them is filled with an oil of viscosity 1 poise. The upper plate of area 1.5 m^2 is required to move with a speed of 0.5 m/s relative to the lower plate. Determine the necessary force and power required to maintain this speed. [03]
- Q - 2 (a) Give the detailed classification of mouthpiece. [02]
- (b) Explain Non-uniform flow, Incompressible flow and Rotational flow [03]
- (c) Check whether the following sets of velocity components satisfy the continuity equation of steady, incompressible flow or not: [04]
- (a) $u = x^2 - y^2$, $v = x - 2xy$
- (b) $u = -\ln xy$, $v = y/x$
- (d) Water flows over a rectangular weir of width 2 m at a depth of 20 cm. Afterwards, water passes through a triangular right-angled weir. Find the depth of water through the triangle weir. The coefficients of discharge for rectangular and triangular weirs are 0.63 and 0.59, respectively. [05]

OR

- Q - 2 (a) The diameter of a pipe at the sections 1-1 and 2-2 are 150 mm and 250 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 4.5m/s. Calculate (i) discharge through the pipe, and (ii) velocity of water at section 2-2. [02]
- (b) A convergent-divergent mouthpiece is fitted to the vertical side of a tank containing water. The throat and exit diameters are 6 cm and 10 cm respectively. If the maximum pressure head at throat is 2.5 m of water absolute, find the maximum head of water [03]

for steady flow. Atmospheric pressure head = 10.3 m of water.

(c) Derive an expression for discharge through a fully submerged orifice. [04]

(d) Derive the continuity equation for three dimensional flow. [05]

Q - 3 (a) Define Free vortex flow and Forced vortex flow. [02]

(b) Water flows through a 300 mm x 150 mm venturimeter at the rate of $0.065 \text{ m}^3/\text{s}$ and the differential gauge is deflected 1.2 m. Specific gravity of the manometric liquid is 1.6. Determine the coefficient of discharge of the venturimeter. [03]

(c) A 90 mm diameter orifice discharges 30 lit/sec of water under a constant head of 2.5m. A flat plate held normal to the jet just downstream from the orifice requires a force of 200 N to resist the impact of the jet. Calculate the hydraulic co-efficients. [04]

(d) Explain the Lagrangian method for motion of fluid particles. [05]

OR

Q - 3 (a) A stream function is given by $\psi = 8x - 10y$. Determine the velocity components and also magnitude and direction of the resultant velocity at any point. [03]

(b) Determine the time required to lower the water level from 4 m to 3 m in a reservoir of dimensions 80 m x 80 m, by (i) a rectangular notch of length 1 m, and (ii) a right angled V-notch. Take $C_d = 0.62$ [03]

(c) Explain Flow net in detail. [04]

(d) Give the detailed classification of Notches and Weirs. [04]

SECTION - II

Q - 4 (a) Enlist the assumptions made in the derivation of Bernoulli's equation. [02]

(b) State Euler's equation of motion along a streamline. [05]

Water flows through a tapering pipe having diameter 10 cm and 20 cm at section 1 and 2 respectively. The heights of section 1 and 2 above horizontal datum are 3 m and 5 m respectively. The pressure at section 1 is 30 kN/m^2 . The discharge through pipe is $0.03 \text{ m}^3/\text{s}$. Estimate the pressure at section 2.

Q - 5 (a) Explain momentum correction factor. [02]

(b) A rectangular plate of 2 m wide is submerged in water vertically such that its top and bottom surfaces are 2 m and 5 m respectively below the free surface. Determine the total pressure force on one side of the plate and the location of centre of pressure. [03]

(c) Explain Pitot tube. [04]

In a duct of 400 mm diameter, a pitot-static tube is placed in the centre. The mean velocity in the duct is 0.85 of the central velocity. Determine the discharge through the duct if the difference between the static and total pressure is 80 mm of water. Take

the co-efficient of Pitot tube as $C_v = 0.98$.

- (d) State and prove the Hydrostatic law. [05]

OR

- Q - 5 (a) Define Metacentre, Metacentric height, Buoyancy and Centre of pressure [04]

- (b) A cylindrical block weighs 22 kN having diameter of 2 m and height 2.5 m is to float in sea water ($S = 1.025$). Show that it does not float vertically. [04]

- (c) Derive Euler's equation of motion. [06]

- Q - 6 (a) Define Manometer. Explain piezometer in detail with neat sketch. [04]

- (b) Derive an expression for power transmission through pipe. Also find the maximum transmission of power and corresponding efficiency. [05]

- (c) Determine the head loss due to friction in a pipe of diameter 250 mm and length 60 m through which water is flowing at a velocity of 2.5 m/s using (i) Darcy's formula, (ii) Chezy's formula for which $C = 60$. Assume kinematic viscosity for water as 0.01 stoke. [05]

OR

- Q - 6 (a) A U-tube manometer contains the mercury as manometric liquid, and is used to measure vacuum pressure of fluid ($S = 0.8$) in pipe. The other end of the manometer is open to atmosphere. The height of fluid in pipe of the left limb is 30 cm below the centre of pipe. Calculate pressure in the pipe when the difference of mercury level in the two limbs is 60 cm. [04]

- (b) Discuss Total energy line and Hydraulic gradient line with neat sketch. [05]

- (c) A 10 cm diameter pipe is suddenly enlarged to 15 cm diameter pipe. The velocity and pressure of smaller diameter pipe are 5.5 m/s and 130 kN/m² respectively. Determine the loss of head and pressure in the larger pipe due to sudden enlargement. [05]
